



$$\underbrace{O_t, O_{t-1}, O_{t-2}, \dots, O_{t-h}}_{S_t}$$

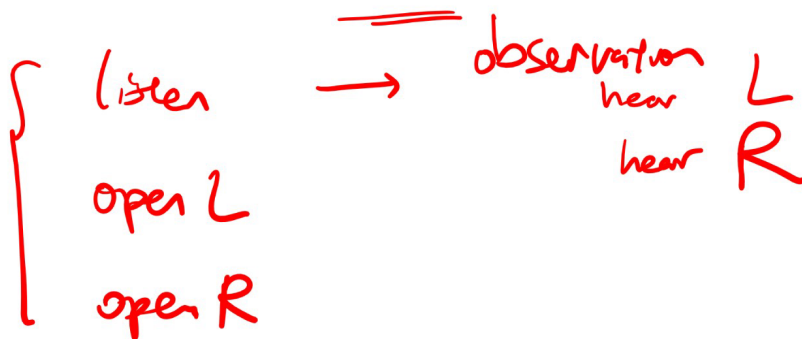
$$\pi_t(O_t | O_{t-1}, \dots, O_0)$$

$$\pi_t(O_t, O_{t-1}, \dots, O_{t-h})$$

finite memory

RNN.

POMDP tiger game



POMDP is NOT Markov

$\hookrightarrow$ belief state MDP

$$\pi(o_t, o_{t+1}, \dots)$$

$$\pi(b_t)$$

prob distr. over state.

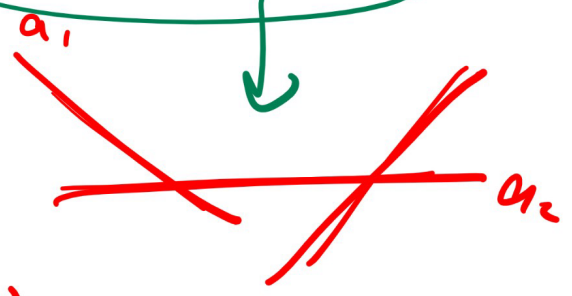
$$\pi(\text{continuous state})$$

$\rightarrow V(b_t)$ is piecewise linear in b_t .

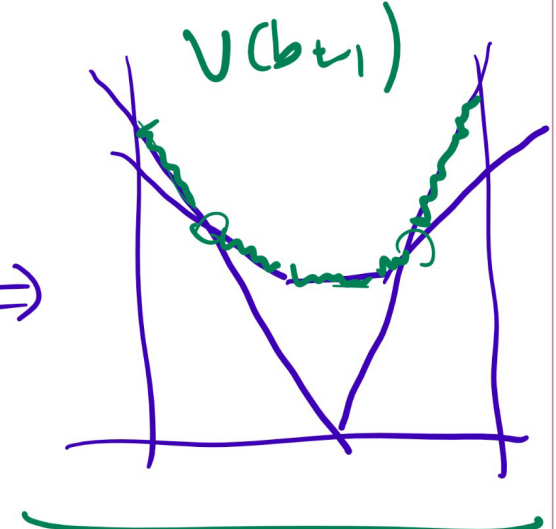
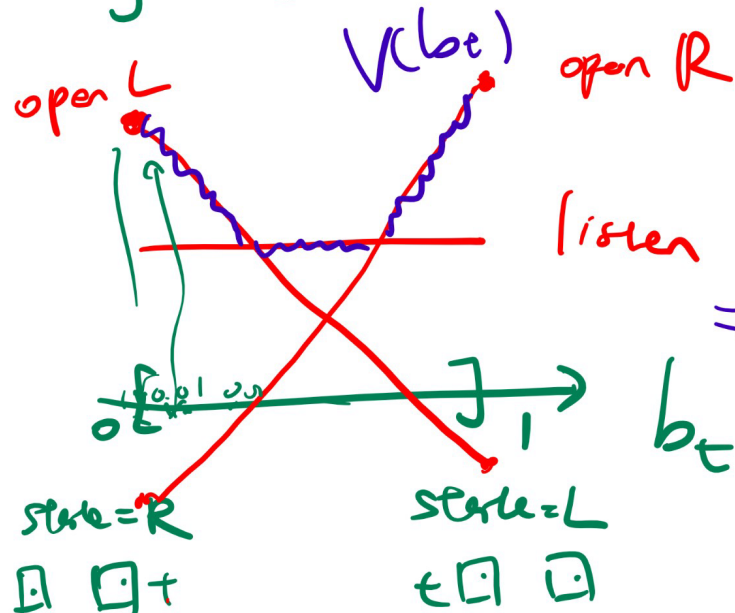
$$\underline{b_t} \cdot \underline{R_t}$$

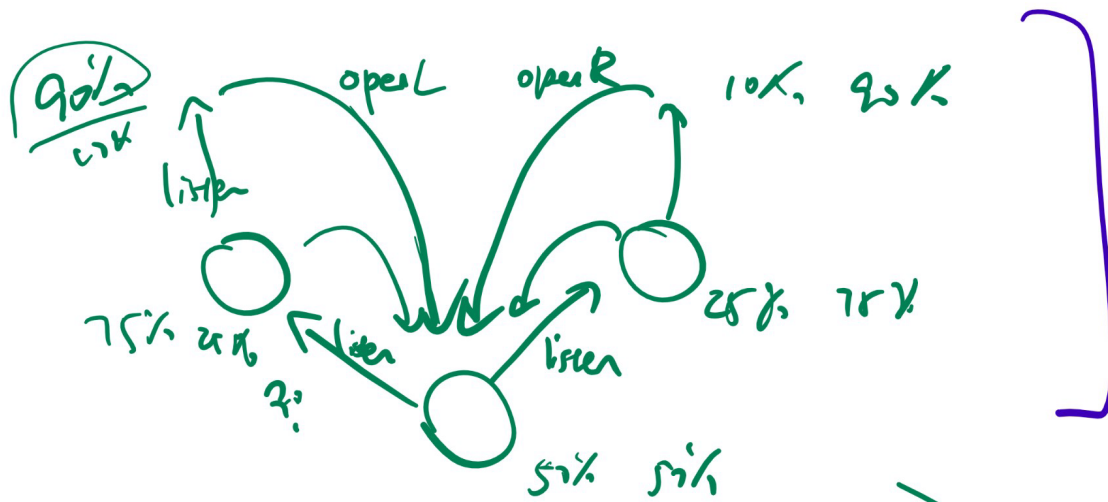
$$\max \underline{Q(a, b_t)}$$

constant



tiger game





- ① IGDS
- ② Support

